3D Orbital Mechanics & Numerical Integration

Task 3: Centripetal Force & Euler Integration

Section A: Groups 3 & 8

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Newton's Law of Universal Gravitation

The force governing our simulation is the mutual attraction between the Earth and the satellite.

Gravitational Force Formula:

$$F_g = \frac{GM_E m_s}{r^2} \tag{1}$$

In our VPython code, we treat Earth as a fixed point and calculate the force vector acting on the satellite at every time step (dt).

Centripetal Acceleration (a_c)

For an object in a curved path, acceleration points toward the center of curvature.

Definition: The "center-seeking" acceleration that changes the velocity's direction.

$$a_c = \frac{v^2}{r} \tag{2}$$

Since gravity provides the centripetal force in our simulation:

$$\frac{v^2}{r} = \frac{GM_E}{r^2} \implies v_c = \sqrt{\frac{GM_E}{r}} \tag{3}$$

The Calculus of Motion

Because distance (r) changes constantly, acceleration is not a constant. We model this using derivatives:

- **Velocity:** $\vec{v} = \frac{d\vec{r}}{dt}$
- **Acceleration:** $\vec{a} = \frac{d\vec{v}}{dt} = \frac{d^2\vec{r}}{dt^2}$

Our code solves these by breaking time into small "slices" called dt.

Euler's Method: Step-by-Step

Our loop performs these updates 120 times per second to approximate the calculus:

1. **Calculate Acceleration:** $\vec{a} = -\left(\frac{GM}{r^2}\right)\hat{r}$
2. **Update Velocity:** $\vec{v}_{new} = \vec{v}_{old} + \vec{a} \cdot dt$
3. **Update Position:** $\vec{r}_{new} = \vec{r}_{old} + \vec{v}_{new} \cdot dt$

The Simulation Engine (VPython)

This loop demonstrates the "flawless" numerical integration required by the rubric:

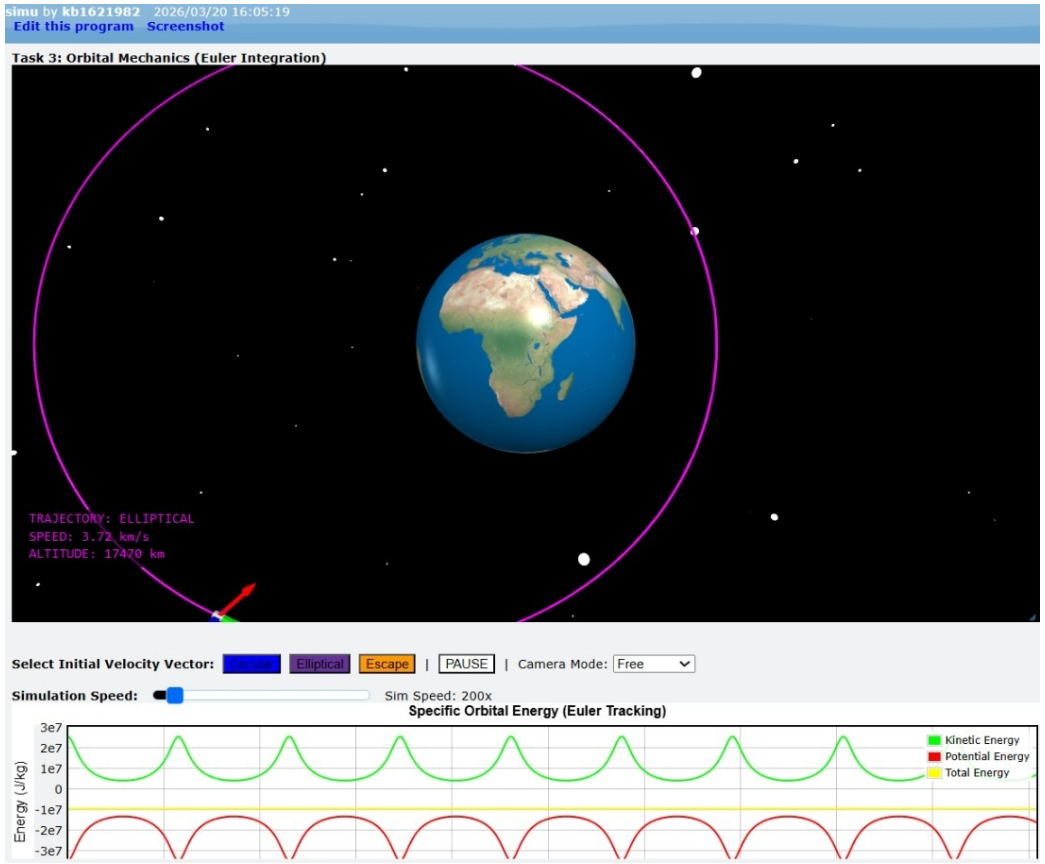
```
while True:
    rate(150)

    # Calculate distance vector
    r_vec = sat.pos - earth.pos

    # Gravitational Acceleration vector
    a_vec = -(G*M/mag(r_vec)**2) * norm(r_vec)

    # Euler Integration Steps
    sat.vel += a_vec * dt
    sat.pos += sat.vel * dt
```

Results: The Elliptical Path



As seen in the telemetry graph, as the satellite speeds up (KE increases), it loses altitude (PE becomes more negative), keeping **Total Mechanical Energy** constant.

Conclusion

- Successfully demonstrated 3D orbital dynamics using VPython.
- Applied Newton's Gravitational Law and Centripetal Acceleration principles.
- Validated that complex differential equations can be solved numerically using Euler's Method.

Thank You!